

NEET (UG) 2025

National Eligibility cum Entrance Test (Undergraduate)

Date: **04 May 2025** | Test Booklet Code: **45**

Duration: 3 hours | 180 MCQs | Maximum Marks: 720

Important Instructions:

1. The test contains **180** multiple-choice questions from **Physics, Chemistry, and Biology (Botany & Zoology)**. All questions are compulsory.
2. Each correct response: **+4 marks**. Each incorrect response: **–1 mark**. Maximum marks: **720**.
3. Use a Blue/Black ballpoint pen only. Use of an electronic/manual calculator is prohibited.
4. The official answer key is provided at the end of this booklet.

Disclaimer: This document is a faithful reproduction of the question paper for educational and revision purposes. For the official paper and answer key, please refer to NTA / NMC.

PHYSICS (Questions 1–46)

Q1. Consider a water tank with one wall at $x = L$, very wide in z direction. Filled with liquid of surface tension S and density ρ , the liquid surface makes angle θ with x -axis at $x = L$. If $y(x)$ is the height of the surface, the equation for $y(x)$ is (take $\theta \approx \sin \theta \approx \tan \theta = dy/dx$, g is acceleration due to gravity):

- (1) $d^2y/dx^2 = (\rho g/S)x$
- (2) $d^2y/dx^2 = (\rho g/S)y$
- (3) $d^2y/dx^2 = \sqrt{(\rho g/S)}$
- (4) $dy/dx = \sqrt{(\rho g/S)}x$

Q2. A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is:

- (1) 100
- (2) 125
- (3) 150
- (4) 250

Q3. An electron (mass 9×10^{-31} kg, charge 1.6×10^{-19} C) moving with speed $c/100$ is injected into a magnetic field B of magnitude 9×10^{-4} T perpendicular to its motion. We wish to apply a uniform electric field E together with B so the electron does not deflect. Then ($c = 3 \times 10^8$ m/s):

- (1) $E \perp B$ and $|E| = 27 \times 10^4$ V/m
- (2) $E \perp B$ and $|E| = 27 \times 10^2$ V/m
- (3) $E \parallel B$ and $|E| = 27 \times 10^2$ V/m
- (4) $E \parallel B$ and $|E| = 27 \times 10^4$ V/m

Q4. Two inclined surfaces of equal length L and same angle 45° with horizontal: one rough, the other perfectly smooth. A body takes 2 times as much time to slide down the rough surface as on the smooth surface. The coefficient of kinetic friction (μ) between the object and the rough surface is close to:

- (1) 0.25
- (2) 0.40
- (3) 0.5
- (4) 0.75

Q5. Kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying brakes, A stops after 1000 m and B stops after 1500 m. If F_A and F_B are the braking forces on A and B respectively, the ratio F_A/F_B is:

- (1) $3/2$
- (2) $2/3$
- (3) $1/3$
- (4) $1/2$

Q6. The current passing through the battery in the given circuit (Wheatstone bridge with 5 V battery, $1/3 \Omega$, 5.5Ω external) is:

- (1) 2.0 A
- (2) 0.5 A
- (3) 2.5 A
- (4) 1.5 A

Q7. A bob of heavy mass m is suspended by a light string of length l . The bob is given a horizontal velocity v . If the string gets slack at point P making angle θ from the horizontal, the ratio of the speed v at P to its initial speed v is:

- (1) $(\sin \theta)^{1/2}$
- (2) $[1/(2 + 3 \sin \theta)]^{1/2}$
- (3) $[\cos \theta/(2 + 3 \sin \theta)]^{1/2}$
- (4) $[\sin \theta/(2 + 3 \sin \theta)]^{1/2}$

Q8. The output Y of the given logic implementation (with inputs A, B and gates as shown) is similar to the output of a/an _____ gate.

- (1) AND
- (2) NAND
- (3) OR
- (4) NOR

Q9. The electric field in a plane EM wave is $E_z = 60 \cos(5x + 1.5 \times 10^8 t)$ V/m. The expression for the corresponding magnetic field is (subscripts denote direction):

- (1) $B_y = 2 \times 10^{-8} \cos(5x + 1.5 \times 10^8 t)$ T
- (2) $B_x = 2 \times 10^{-8} \cos(5x + 1.5 \times 10^8 t)$ T
- (3) $B_z = 60 \cos(5x + 1.5 \times 10^8 t)$ T
- (4) $B_y = 60 \sin(5x + 1.5 \times 10^8 t)$ T

Q10. A 0.5 kg ball is dropped from 40 m. It rises to 10 m after the bounce. The impulse imparted to the ball during collision with the ground is ($g = 9.8 \text{ m/s}^2$):

- (1) 21 Ns
- (2) 7 Ns
- (3) 0
- (4) 84 Ns

Q11. AB is a part of an electrical circuit (1 H inductor, 5 V battery, 2Ω resistor in series). The potential difference $V_A - V_B$ at the instant when current $i = 2$ A and is increasing at 1 A/s is:

- (1) 5 volt
- (2) 6 volt
- (3) 9 volt
- (4) 10 volt

Q12. A 2 A current flows through two different small circular copper coils with radii ratio 1:2. The ratio of their respective magnetic moments is:

- (1) 1:4
- (2) 1:2
- (3) 2:1
- (4) 4:1

Q13. A combination of four similar thin convex lenses arranged axially in contact. The power and total magnification, compared to power p and magnification m of each lens, are:

- (1) $4p$ and $4m$
- (2) p and $4m$
- (3) $4p$ and m
- (4) p and m

Q14. An oxygen cylinder of volume 30 L has 18.20 mol of O_2 . After some O_2 is withdrawn, gauge pressure drops to 11 atm at 27 °C. The mass withdrawn is nearly ($R = 100/12 \text{ J mol}^{-1} \text{ K}^{-1}$, $M(O_2) = 32$, 1 atm = $1.01 \times 10^5 \text{ N/m}^2$):

- (1) 0.125 kg
- (2) 0.144 kg
- (3) 0.116 kg
- (4) 0.156 kg

Q15. Time and position of a moving particle are related by $t = x^2 + x$. The acceleration of the particle is:

- (1) $-2/(x + 2)^3$
- (2) $-2/(2x + 1)^3$
- (3) $+2/(x + 1)^3$
- (4) $2/(2x + 1)$

Q16. An AC supply of 220 V at 50 Hz is connected in series with $R = 20 \Omega$, capacitor of reactance 25Ω and inductor of reactance 45Ω . The current and phase angle between current and voltage are:

- (1) 7.8 A and 30°
- (2) 7.8 A and 45°
- (3) 15.6 A and 30°
- (4) 15.6 A and 45°

Q17. The Sun rotates around its centre once in 27 days. What is the period if it expanded to twice its present radius without external influence (uniform density sphere)?

- (1) 100 days
- (2) 105 days
- (3) 115 days
- (4) 108 days

Q18. A model for quantized motion of an electron in a uniform magnetic field B states that the flux through the orbit is $n(h/e)$. The magnetic moment in the lowest energy state is (m = electron mass):

- (1) $he/\pi m$
- (2) $he/(2\pi m)$
- (3) $heB/\pi m$
- (4) $heB/(2\pi m)$

Q19. Three identical heat-conducting rods are connected in series. Side rods have thermal conductivity $2K$, middle has K . Left end at $3T$, right end at T (insulated outside). In steady state, junctions T_1 (left) and T_2 (right). The ratio T_1/T_2 is:

- (1) $3/2$
- (2) $4/3$
- (3) $5/3$
- (4) $5/4$

Q20. Plates of a parallel plate capacitor are separated by d . Two slabs of dielectric K_1 and K_2 with thicknesses $3d/8$ and $d/2$ respectively are inserted. Capacitance becomes $2\times$ the air value. If $K_1 = 1.25 K_2$, the value of K_2 is:

- (1) 2.66
- (2) 2.33
- (3) 1.60
- (4) 1.33

Q21. Two cities X and Y are connected by a regular bus service with a bus leaving every T min in either direction. A girl driving at 60 km/h from X to Y notices a bus passes her every 30 min in her direction and every 10 min in the opposite. The period T and bus speed are:

- (1) 9 min, 40 km/h
- (2) 25 min, 100 km/h
- (3) 10 min, 90 km/h
- (4) 15 min, 120 km/h

Q22. A uniform rod (mass 20 kg, length 5 m) leans against a smooth vertical wall at 60° with it. Other end on rough horizontal floor. The friction force from the floor is ($g = 10 \text{ m/s}^2$):

- (1) 100 N
- (2) $100\sqrt{3}$ N
- (3) 200 N
- (4) $200\sqrt{3}$ N

Q23. In an oscillating spring–mass system, sand leaks slowly from the box. Which option correctly schematically depicts the changes in average frequency $\omega(t)$ and amplitude $A(t)$?

- (1) ω increases, A constant
- (2) ω increases, A decreases
- (3) ω decreases, A increases
- (4) ω decreases, A constant

Q24. A balloon of surface tension S, outlet area A, gas density ρ , takes spherical shape of radius R. When gas flows out freely, r changes from R to 0 in time T. If $v(r) \propto r^a$ and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$, then:

- (1) $a = 1/2, \alpha = 1/2, \beta = -1, \gamma = +1, \delta = 3/2$
- (2) $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = -1/2, \delta = 5/2$
- (3) $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = 1/2, \delta = 7/2$
- (4) $a = 1/2, \alpha = 1/2, \beta = -1/2, \gamma = 1/2, \delta = 7/2$

Q25. A spherical object's diameter measured with vernier callipers: 10 VSD = 9 MSD, least MSD division = 0.1 cm, zero of VS at $x = 0.1$ cm when jaws are closed. MSR = 5 cm, coinciding vernier division = 8. Measured diameter after zero error correction is:

- (1) 5.18 cm
- (2) 5.08 cm
- (3) 4.98 cm
- (4) 5.00 cm

Q26. A parallel plate capacitor (circular plates) is being charged so that surface charge density increases at a constant rate. The magnetic field due to displacement current is:

- (1) zero everywhere
- (2) constant between plates and zero outside
- (3) non-zero everywhere with maximum at the imaginary cylindrical surface joining peripheries
- (4) zero between plates and non-zero outside

Q27. Unpolarized light in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then:

- (1) reflected light is completely polarized and angle of reflection $\approx 60^\circ$
- (2) reflected light is partially polarized and angle of reflection $\approx 30^\circ$
- (3) both reflected and transmitted are perfectly polarized with angles $\approx 60^\circ$ and 30°
- (4) transmitted light is completely polarized with angle of refraction $\approx 30^\circ$

Q28. Two identical charged conducting spheres A and B, charge q each, repel with force F . A third identical uncharged sphere is brought in contact with A first, then with B, then removed. The new force of repulsion between A and B is:

- (1) $3F/5$
- (2) $2F/3$
- (3) $F/2$
- (4) $3F/8$

Q29. A container has two chambers $V_1 = 2 \text{ L}$ and $V_2 = 3 \text{ L}$ separated by a thermal insulator partition. Chambers contain $n_1 = 5 \text{ mol}$ and $n_2 = 4 \text{ mol}$ of ideal gas at $p_1 = 1 \text{ atm}$ and $p_2 = 2 \text{ atm}$. When partition is removed, equilibrium pressure is:

- (1) 1.3 atm
- (2) 1.6 atm
- (3) 1.4 atm
- (4) 1.8 atm

Q30. A particle of mass m moves around the origin with a constant inward force F . If Bohr's model is applied, radius r and speed v of the n -th orbit depend on n as:

- (1) $r \propto n^{1/3}$; $v \propto n^{1/3}$
- (2) $r \propto n^{1/3}$; $v \propto n^{2/3}$
- (3) $r \propto n^{2/3}$; $v \propto n^{1/3}$
- (4) $r \propto n^{4/3}$; $v \propto n^{-1/3}$

Q31. The radius of Mars's orbit is about $4\times$ that of Mercury. Martian year is 687 Earth days. The length of one year on Mercury is:

- (1) 88 earth days
- (2) 225 earth days
- (3) 172 earth days
- (4) 124 earth days

Q32. A body weighs 48 N on Earth's surface. Its gravitational force at a height equal to one-third the Earth's radius from its surface is:

- (1) 16 N
- (2) 27 N
- (3) 32 N
- (4) 36 N

Q33. A wire of resistance R is cut into 8 equal pieces. From these, two equivalent resistances are made by connecting four pieces in parallel. The two sets are then connected in series. The net effective resistance is:

- (1) $R/64$
- (2) $R/32$
- (3) $R/16$
- (4) $R/8$

Q34. De-Broglie wavelength of an electron orbiting in $n = 2$ of hydrogen is close to (Bohr radius = 0.052 nm):

- (1) 0.067 nm
- (2) 0.67 nm
- (3) 1.67 nm
- (4) 2.67 nm

Q35. An electric dipole ($5 \times 10^{-18} \text{ C}\cdot\text{m}$) is aligned with a uniform field of $4 \times 10^5 \text{ N/C}$. It is rotated through 60° w.r.t. the field. The change in potential energy is:

- (1) 0.8 J
- (2) 1.0 J
- (3) 1.2 J
- (4) 1.5 J

Q36. A constant 50 V is maintained between A and B of the circuit (with 1Ω , 2Ω , 3Ω , 4Ω as shown). The current through branch CD is:

- (1) 1.5 A
- (2) 2.0 A
- (3) 2.5 A
- (4) 3.0 A

Q37. A photon and an electron of mass m have the same energy E . The ratio $\lambda_{\text{photon}}/\lambda_{\text{electron}}$ is (c is speed of light):

- (1) $\sqrt{E/2m}$
- (2) $c\sqrt{2mE}$
- (3) $c\sqrt{2m/E}$
- (4) $(1/c)\sqrt{E/2m}$

Q38. Which option(s) represent the variation of photoelectric current with the property of light shown on the x-axis (A: vs intensity-linear; B: vs intensity-flat; C: vs frequency-linear; D: vs frequency-linear)?

- (1) A only
- (2) A and C
- (3) A and D
- (4) B and D

Q39. A sphere of radius R is cut from a larger solid sphere of radius $2R$. The ratio of moment of inertia of the smaller sphere to the rest about the Y-axis is:

- (1) $7/8$
- (2) $7/40$
- (3) $7/57$
- (4) $7/64$

Q40. A full-wave rectifier with diodes D_1 and D_2 . If $V_{\text{in}} = 220 \sin(100\pi t) \text{ V}$, then at $t = 15 \text{ ms}$:

- (1) D_1 forward, D_2 reverse
- (2) D_1 reverse, D_2 forward
- (3) Both forward
- (4) Both reverse

Q41. Two gases A and B at the same pressure in cylinders with movable pistons of radii r_A and r_B . Equal heat is supplied reversibly at constant pressure; pistons of A and B move 16 cm and 9 cm. If ΔU is same, the ratio r_A/r_B is:

- (1) $4/3$
- (2) $3/4$
- (3) $2/\sqrt{3}$
- (4) $\sqrt{3}/2$

Q42. A physical quantity $P = a^3b^2/(c\sqrt{d})$. Percent errors in a, b, c, d are 1%, 3%, 2%, 4%. The percent error in P is:

- (1) 10%
- (2) 2%
- (3) 13%
- (4) 15%

Q43. A polaroid is placed at 22.5° between two crossed polarizers. The intensity of transmitted light ($I =$ intensity after first polaroid) is:

- (1) $I/2$
- (2) $I/4$
- (3) $I/8$
- (4) $I/16$

Q44. Two identical point masses P and Q on springs of constants k_1 and k_2 oscillate vertically. Maximum speeds are equal. The ratio A_Q/A_P is:

- (1) k_1/k_2
- (2) k_2/k_1
- (3) $\sqrt{(k_1/k_2)}$
- (4) $\sqrt{(k_2/k_1)}$

Q45. A pipe open at both ends has fundamental frequency f . Dipped vertically in water to half its length, the new fundamental frequency is:

- (1) $f/2$
- (2) f
- (3) $3f/2$
- (4) $2f$

Q46. Ratio of wavelengths of light absorbed by hydrogen for $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions is:

- (1) $1/36$
- (2) $1/16$
- (3) $1/9$
- (4) $1/4$

CHEMISTRY (Questions 47–90)

Q47. Which of the following statements are true? A. Unlike Ga (high m.p.), Cs has very low m.p. B. Electronegativity values of N and Cl on Pauling scale are not the same. C. Ar, K⁺, Cl⁻, Ca²⁺, S²⁻ are all isoelectronic. D. First IE order: Si > Al > Mg > Na. E. Atomic radius of Cs is greater than Li and Rb.

- (1) A, B and E only
- (2) C and E only
- (3) C and D only
- (4) A, C and E only

Q48. Match (Ion) with Group in cation analysis: A. Co²⁺ B. Mg²⁺ C. Pb²⁺ D. Al³⁺ ↔ I. Group I II. Group III III. Group IV IV. Group VI

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-II, C-IV, D-I
- (4) A-III, B-II, C-I, D-IV

Q49. Predict major product P: cyclopent-1-ene with CH₃NH₂ → (i) HBr, benzoyl peroxide; (ii) KCN; (iii) Na(Hg)/C₆H₅OH

- (1) 1-(aminomethyl)-2-methylcyclopentane
- (2) 1-(aminomethyl)-1-methylcyclopentane
- (3) 1-(cyano)-2-methylcyclopentane
- (4) 1-(cyano)-1-methylcyclopentane

Q50. Energy and radius of first Bohr orbit of He⁺ and Li²⁺ ($R_H = 2.18 \times 10^{-18}$ J, $a_0 = 52.9$ pm):

- (1) $E(\text{Li}^{2+}) = -19.62 \times 10^{-18}$ J; $r = 17.6$ pm; $E(\text{He}^+) = -8.72 \times 10^{-18}$ J; $r = 26.4$ pm
- (2) $E(\text{Li}^{2+}) = -8.72 \times 10^{-18}$ J; $r = 26.4$ pm; $E(\text{He}^+) = -19.62 \times 10^{-18}$ J; $r = 17.6$ pm
- (3) $E(\text{Li}^{2+}) = -19.62 \times 10^{-18}$ J; $r = 17.6$ pm; $E(\text{He}^+) = -8.72 \times 10^{-18}$ J; $r = 26.4$ pm
- (4) $E(\text{Li}^{2+}) = -8.72 \times 10^{-18}$ J; $r = 17.6$ pm; $E(\text{He}^+) = -19.62 \times 10^{-18}$ J; $r = 17.6$ pm

Q51. Which of the following are paramagnetic? A. [NiCl₄]²⁻ B. Ni(CO)₄ C. [Ni(CN)₄]²⁻ D. [Ni(H₂O)₆]²⁺ E. Ni(PPh₃)₄

- (1) A and C only
- (2) B and E only
- (3) A and D only
- (4) A, D and E only

Q52. Statement I: Like nitrogen forms ammonia, arsenic forms arsine. Statement II: Antimony cannot form antimony pentoxide.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q53. Which configurations belong to main-group elements? A. [Ne]3s¹ B. [Ar]3d³4s² C. [Kr]4d¹5s²5p¹ D. [Ar]3d¹4s¹ E. [Rn]5f¹⁴6d²7s²

- (1) B and E only
- (2) A and C only
- (3) D and E only
- (4) A, C and D only

Q54. Dalton's atomic theory could NOT explain which?

- (1) Law of conservation of mass
- (2) Law of constant proportion
- (3) Law of multiple proportion
- (4) Law of gaseous volume

Q55. Oxidation states of underlined K in $\text{KO}\underline{\text{O}}$, O in $\text{H}\underline{\text{O}}\text{O}$, S in $\text{H}\underline{\text{S}}\text{O}$ respectively:

- (1) +1, -1, and +6
- (2) +2, -2, and +6
- (3) +1, -2, and +4
- (4) +4, -4, and +6

Q56. If $t_{1/2}$ for a first-order reaction is 1 minute, time for 99.9% completion is closest to:

- (1) 2 minutes
- (2) 4 minutes
- (3) 5 minutes
- (4) 10 minutes

Q57. Correct order of wavelength absorbed by: A. $[\text{Co}(\text{NH}_3)_6]^{3+}$ B. $[\text{Co}(\text{CN})_6]^{3-}$ C. $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$

- (1) $B < D < A < C$
- (2) $B < A < D < C$
- (3) $C < D < A < B$
- (4) $C < A < D < B$

Q58. Which compound exhibits cis-trans isomers?

- (1) Pent-1-ene
- (2) 2-Methylhex-2-ene
- (3) 1,1-Dimethylcyclopropane
- (4) 1,2-Dimethylcyclohexane

Q59. For phosphoric acid (K_1 , K_2 , K_3 , K = overall), which are true? A. $\log K = \log K_1 + \log K_2 + \log K_3$ B. H_3PO_4 stronger than H_2PO_4^- and HPO_4^{2-} C. $K_1 > K_2 > K_3$ D. $K_1 = (K_2 + K_3)/2$

- (1) A and B only
- (2) A and C only
- (3) B, C and D only
- (4) A, B and C only

Q60. Which reaction does NOT give benzene as product?

- (1) $\text{C}_6\text{H}_5\text{COONa} + \text{sodalime}/\Delta$
- (2) n-hexane + MoO_3 at 773 K, 10–20 atm
- (3) $\text{HC}\equiv\text{CH}$ passed through red-hot iron tube at 873 K
- (4) $\text{C}_6\text{H}_5\text{N}_2\text{Cl} + \text{H}_2\text{O}$ (warm)

Q61. Molar conductivity Λ_m of 0.050 mol/L solution of a monobasic weak acid is $90 \text{ S cm}^2 \text{ mol}^{-1}$. $\Lambda_m^\circ = 349.6$, $\Lambda_m^\circ = 50.4 \text{ S cm}^2 \text{ mol}^{-1}$. Degree of dissociation α is:

- (1) 0.115
- (2) 0.125
- (3) 0.225
- (4) 0.215

Q62. Statement I: A hypothetical diatomic molecule with bond order zero is quite stable. Statement II: As bond order increases, bond length increases.

- (1) Both I and II true
- (2) Both I and II false
- (3) I true, II false
- (4) I true, II true

Q63. Which complex has minimum conductance in solution?

- (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
- (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]$
- (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2$

Q64. Match: A. XeO_3 B. XeF_4 C. XeOF_4 D. XeF_6 $\leftrightarrow$ I. sp^3d (linear) II. sp^3 (pyramidal) III. sp^3d^3 (distorted octahedral) IV. sp^3d^2 (square pyramidal)

- (1) A-II, B-I, C-IV, D-III
- (2) A-II, B-I, C-III, D-IV
- (3) A-IV, B-II, C-III, D-I
- (4) A-IV, B-II, C-I, D-III

Q65. $\text{C(s)} + 2 \text{H}_2\text{(g)} \rightarrow \text{CH}_4\text{(g)}$; $\Delta H = -74.8 \text{ kJ/mol}$. Which energy diagram is accurate (R $\rightarrow$ P)?

- (1) Reactants higher; activation peak; products 74.8 kJ below R
- (2) Products higher than reactants by 74.8 kJ
- (3) Reactants and products at same level
- (4) Activation peak smaller than 74.8 kJ

Q66. Match (Example): A. Humidity B. Alloys C. Amalgams D. Smoke $\leftrightarrow$ (Type): I. Solid in solid II. Liquid in gas III. Solid in gas IV. Liquid in solid

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-II, C-I, D-IV

Q67. Decreasing basic strength among amines:

- (1) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine
- (2) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline
- (3) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine
- (4) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine

Q68. Which have equal number of atoms? A. 212 g Na_2CO_3 B. 248 g Na_2O C. 240 g NaOH D. 12 g H_2 E. 220 g CO_2

- (1) A, B and C only
- (2) A, B and D only
- (3) B, C and D only
- (4) B, D and E only

Q69. Match Vitamin with deficiency disease: A. B₁₂ B. D C. B₆ D. B₉ ↔ I. Cheilosis II. Convulsions III. Rickets IV. Pernicious anaemia

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-IV, B-III, C-II, D-I

Q70. Decreasing acidity of aliphatic acids:

- (1) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
- (2) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$
- (3) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
- (4) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$

Q71. Statement I: Ferromagnetism is an extreme form of paramagnetism. Statement II: Number of unpaired electrons in Cr^{2+} (Z=24) is the same as in Nd^{3+} (Z=60).

- (1) Both I and II true
- (2) Both I and II false
- (3) I true, II false
- (4) I false, II true

Q72. Match (Mixture) with method: A. $\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$ B. Crude oil C. Glycerol from spent-lye D. Aniline–water ↔ I. Distillation under reduced pressure II. Steam distillation III. Fractional distillation IV. Simple distillation

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

Q73. For $\text{A(g)} \rightleftharpoons 2 \text{B(g)}$, backward rate constant is $2500\times$ the forward at 1000 K ($R = 0.0831 \text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$). K_p at 1000 K is:

- (1) 83.1
- (2) 2.077×10^{-3}
- (3) 0.033
- (4) 0.021

Q74. Statement I: Benzenediazonium salt is prepared by aniline + HNO_3 at 273–278 K; decomposes easily in dry state. Statement II: Iodine insertion into benzene is difficult; iodobenzene is prepared via diazonium + KI.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q75. Number of products (incl. stereoisomers) from monochlorination of $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}_3$:

- (1) 2
- (2) 3
- (3) 5
- (4) 6

Q76. Bond dissociation energy order of starred C–H bond in: I (benzene C–H), II (alkyne C–H), III (cyclopropene C–H):

- (1) II > I > III
- (2) I > II > III
- (3) III > II > I
- (4) II > III > I

Q77. Which compound does NOT decolourize bromine water?

- (1) Cyclohexane
- (2) Phenol
- (3) Styrene
- (4) Aniline

Q78. Major product: $\text{PhCOCH}_2\text{CH}_2\text{CN} + \text{(i) excess } \text{CH}_3\text{MgBr}, \text{(ii) } \text{H}_2\text{O} \rightarrow$

- (1) Ph-cycle with CH_3 , OH, CN
- (2) $\text{Ph}-\text{C}(\text{CH}_3)(\text{OH})-\text{CH}_2\text{CH}_2-\text{C}(\text{CH}_3)=\text{O}$
- (3) $\text{Ph}-\text{C}(\text{OH})(\text{CH}_3)-\text{CH}_2-\text{C}(\text{CH}_3)_2-\text{OH}$
- (4) $\text{Ph}-\text{CO}-\text{CH}_2-\text{CH}_2-\text{CO}-\text{CH}_3$

Q79. Aqueous solution with highest boiling point:

- (1) 0.01 M Urea
- (2) 0.01 M KNO_3
- (3) 0.01 M Na_2SO_4
- (4) 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6$

Q80. Match: A. Haber process B. Wacker oxidation C. Wilkinson catalyst D. Ziegler catalyst $\leftrightarrow$ I. Fe II. PdCl_2 III. $[(\text{PPh}_3)_3\text{RhCl}]$ IV. $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-II, C-III, D-IV
- (4) A-I, B-IV, C-III, D-II

Q81. 5 mol of liquid X + 10 mol of liquid Y has vapour pressure 70 torr. Pure X = 63 torr, pure Y = 78 torr. Which is true about the solution?

- (1) Positive deviation
- (2) Negative deviation
- (3) Ideal
- (4) Volume > sum of individual volumes

Q82. Sugar X: A. Found in honey. B. Keto sugar. C. Exists in α/β anomers. D. Laevorotatory. X is:

- (1) D-Glucose
- (2) D-Fructose
- (3) Maltose
- (4) Sucrose

Q83. Reagent for the conversion: $\text{PhCOOCH}_3 \rightarrow \text{PhCHO}$

- (1) (i) LiAlH_4 , (ii) $\text{H}_2\text{O}/\text{H}^+$
- (2) (i) $\text{AlH}(\text{iBu})_3$, (ii) H_2O
- (3) (i) NaBH_4 , (ii) $\text{H}_2\text{O}/\text{H}^+$
- (4) (i) $\text{H}_2/\text{Pd}-\text{BaSO}_4$

Q84. Assertion: 1-iodobutane undergoes S_N2 faster than 1-chlorobutane. Reason: Iodine is a better leaving group due to its larger size.

- (1) Both A and R true; R is correct explanation
- (2) Both A and R true; R is not correct explanation
- (3) A true, R false
- (4) A false, R true

Q85. Standard heat of formation of Ba^{2+} in kcal/mol [given $\Delta H^\circ_f(SO_4^{2-} \text{ aq}) = -216$, $\Delta H^\circ_{\text{cryst}}(BaSO_4) = -4.5$, $\Delta H^\circ_f(BaSO_4 \text{ s}) = -349$]:

- (1) -128.5
- (2) -133.0
- (3) +133.0
- (4) +220.5

Q86. Total possible isomers (structural + stereo) of cyclic ethers C_4H_8O :

- (1) 6
- (2) 8
- (3) 10
- (4) 11

Q87. Identify correct orders: A. $H_2O > NH_3 > CHCl_3$ (dipole moment) B. $XeF_4 > XeO_3 > XeF_6$ (lone pairs on central atom) C. $O-H > C-H > N-O$ (bond length) D. $N \equiv O > H \equiv C > H_2$ (bond enthalpy)

- (1) A, D only
- (2) B, D only
- (3) A, C only
- (4) B, C only

Q88. Higher yield of NO in $N_2 + O_2 \rightleftharpoons 2 NO$ ($\Delta H = +180.7 \text{ kJ/mol}$) at: A. higher T B. lower T C. higher $[N_2]$ D. higher $[O_2]$

- (1) A, D only
- (2) B, C only
- (3) B, C, D only
- (4) A, C, D only

Q89. Rate constant 0.03 s^{-1} . Time for 7.2 mol/L to drop to 0.9 mol/L ($\log 2 = 0.301$):

- (1) 69.3 s
- (2) 23.1 s
- (3) 210 s
- (4) 21.0 s

Q90. Which reaction does NOT belong to Lassaigne's test?

- (1) $Na + C + N \rightarrow NaCN$
- (2) $2 Na + S \rightarrow Na_2S$
- (3) $Na + X \rightarrow NaX$
- (4) $2 CuO + C \rightarrow 2 Cu + CO_2$

BIOLOGY (Botany & Zoology) (Questions 91–180)

Q91. Complex II of mitochondrial ETC is also known as:

- (1) Cytochrome bc₁■
- (2) Succinate dehydrogenase
- (3) Cytochrome c oxidase
- (4) NADH dehydrogenase

Q92. PCR amplifies DNA following the equation:

- (1) N^2
- (2) $2^{\frac{n}{2}}$ ■
- (3) $2n + 1$
- (4) $2N^2$

Q93. Potential drawbacks in adoption of IVF: A. High fatality risk to mother B. Expensive instruments and reagents C. Husband/wife necessary as donors D. Less adoption of orphans E. Not available in India F. Possibility that early embryo does not survive

- (1) B, D, F only
- (2) A, C, D, F only
- (3) A, B, C, D only
- (4) A, B, C, E, F only

Q94. Blood vessel that carries deoxygenated blood from body to heart in a frog:

- (1) Aorta
- (2) Pulmonary artery
- (3) Pulmonary vein
- (4) Vena cava

Q95. Reductionist Biology refers to:

- (1) Physico-chemical approach
- (2) Physiological approach
- (3) Chemical approach
- (4) Behavioural approach

Q96. Statement I: In RNA world, RNA was the first genetic material; reactive and unstable. Statement II: DNA evolved from RNA, more stable; complementary strands resist changes via repair mechanisms.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q97. Epiphytes growing on a mango branch is an example of:

- (1) Commensalism
- (2) Mutualism
- (3) Predation
- (4) Amensalism

Q98. Choose correct: A. Eukaryotic ribosomes 80S, prokaryotic 70S. B. Each ribosome has two sub-units. C. $80S = 60S + 40S$; $70S = 50S + 30S$. D. $80S = 60S + 20S$; $70S = 50S + 20S$. E. $80S = 60S + 30S$; $70S = 50S + 30S$.

- (1) A, B, C true
- (2) A, B, D true
- (3) A, B, E true
- (4) B, D, E true

Q99. Example of ex-situ conservation:

- (1) National Park
- (2) Wildlife Sanctuary
- (3) Zoos and botanical gardens
- (4) Protected areas

Q100. Statement I: Solar energy is the primary source of energy in an ecosystem. Statement II: The rate of production of organic matter during photosynthesis is called net primary productivity (NPP).

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q101. Match: A. Emphysema B. Angina Pectoris C. Glomerulonephritis D. Tetany $\leftrightarrow$ I. Spasms (low Ca^{2+}) II. Damaged alveolar walls III. Acute chest pain (low O_2 to heart) IV. Inflammation of glomeruli

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-III, C-IV, D-I

Q102. A: Wind/water-pollinated flowers are not very colourful and lack nectar. R: They produce enormous amounts of pollen.

- (1) A and R true; R is correct explanation
- (2) A and R true; R is NOT correct explanation
- (3) A true, R false
- (4) A false, R true

Q103. Non-distilled alcoholic beverage produced by yeast:

- (1) Whisky
- (2) Brandy
- (3) Beer
- (4) Rum

Q104. Statement I: In a floral formula, $\oplus$ stands for zygomorphic and G stands for inferior ovary. Statement II: $\oplus$ stands for actinomorphic; G stands for superior ovary.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q105. Streptokinase from *Streptococcus* is used for:

- (1) Curd production
- (2) Ethanol production
- (3) Liver disease treatment
- (4) Removing clots from blood vessels

Q106. Chromosome with the highest number of genes in human genome:

- (1) Chromosome X
- (2) Chromosome Y
- (3) Chromosome 1
- (4) Chromosome 10

Q107. Location of the male frog's copulatory pad:

- (1) First and second digit of fore limb
- (2) First digit of hind limb
- (3) Second digit of fore limb
- (4) First digit of fore limb

Q108. Phytohormone that promotes nutrient mobilisation, delaying leaf senescence:

- (1) Ethylene
- (2) Absciscic acid
- (3) Gibberellin
- (4) Cytokinin

Q109. An animal has a body cavity with mesoderm only on the body wall, not surrounding the alimentary canal. Possible coelome:

- (1) Acoelomate
- (2) Pseudocoelomate
- (3) Schizocoelomate
- (4) Spongocoelomate

Q110. Match: A. Head B. Middle piece C. Acrosome D. Tail ↔ I. Enzymes II. Sperm motility III. Energy IV. Genetic material

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-I, D-IV

Q111. Correct sequence in pteridophyte life cycle: A. Prothallus stage B. Meiosis in spore mother cells C. Fertilisation D. Formation of archegonia and antheridia E. Transfer of antherozoids to archegonia in water

- (1) B, A, D, E, C
- (2) B, A, E, C, D
- (3) D, E, C, A, B
- (4) E, D, C, B, A

Q112. Cardiac activities are regulated by: A. Nodal tissue B. A neural centre in medulla oblongata C. Adrenal medullary hormones D. Adrenal cortical hormones

- (1) A, B and C only
- (2) A, B, C and D
- (3) A, C and D only
- (4) A, B and D only

Q113. Which organism cannot fix nitrogen? A. Azotobacter B. Oscillatoria C. Anabaena D. Volvox E. Nostoc

- (1) A only
- (2) D only
- (3) B only
- (4) E only

Q114. Statement I: tRNA and rRNA do not interact with mRNA. Statement II: RNAi takes place in all eukaryotes as cellular defence.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q115. In a plasmid (BamHI, TetR, Ori, EcoRI, ampR, β -galactosidase) — alien DNA inserted at EcoRI. Strategy to select recombinants:

- (1) Use ampicillin & tetracycline plate
- (2) Blue colonies are selected
- (3) White colonies are selected
- (4) Blue colonies grown on ampicillin plates can be selected

Q116. Genetically engineered organism used by Eli Lilly to prepare human insulin:

- (1) Bacterium
- (2) Yeast
- (3) Virus
- (4) Phage

Q117. Class of enzyme that catalyzes: $S-G + S^{\#} \rightarrow S + S^{\#}-G$

- (1) Hydrolase
- (2) Lyase
- (3) Transferase
- (4) Ligase

Q118. Which is NOT correct about monocot stem?

- (1) Hypodermis is parenchymatous
- (2) Vascular bundles are scattered
- (3) Vascular bundles are conjoint and closed
- (4) Phloem parenchyma is absent

Q119. Sequence in bryophyte life cycle: A. Fusion of antherozoid with egg B. Attachment of gametophyte to substratum C. Reduction division for haploid spores D. Formation of sporophyte E. Release of antherozoids into water

- (1) D, E, A, C, B
- (2) B, E, A, C, D
- (3) B, E, A, D, C
- (4) D, E, A, B, C

Q120. Which are correct: A. CT/MRI detect cancers of internal organs. B. Chemotherapeutics kill non-cancerous cells. C. α -interferon activates immune system to destroy tumour. D. Chemotherapeutic drugs are biological response modifiers. E. In leukaemia, blood cell counts are decreased.

- (1) B and D only
- (2) D and E only
- (3) C and D only
- (4) A and C only

Q121. Match: A. Centromere B. Cilium C. Cristae D. Cell membrane $\leftrightarrow$ I. Mitochondrion II. Cell division III. Cell movement IV. Phospholipid bilayer

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-III, C-I, D-IV

Q122. Match: A. Chlorophyll a B. Chlorophyll b C. Xanthophylls D. Carotenoids $\leftrightarrow$ I. Yellow-green II. Yellow III. Blue-green IV. Yellow to yellow-orange

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-I, B-II, C-IV, D-III
- (4) A-I, B-IV, C-III, D-II

Q123. Correct statements: A. In human pregnancy, major organ systems form by end of 12 weeks. B. By end of 8 weeks. C. Heart forms after one month of gestation. D. Limbs and digits develop by end of 2nd month. E. Hair appears in 5th month.

- (1) A and E only
- (2) B and C only
- (3) B, C, D and E only
- (4) A, C, D and E only

Q124. Outer covering of endosperm separates the embryo by a protein-rich layer called:

- (1) Coleoptile
- (2) Coleorhiza
- (3) Integument
- (4) Aleurone layer

Q125. Which diagram is correct for proximal (P) and distal (D) tubule transport of nephron?

- (1) P: HCO_3^- in, NaCl in, H^+/NH_4^+ out; D: NaCl in, H_2O in, HCO_3^- out, K^+/H^+ out (option 1)
- (2) P: $\text{HCO}_3^-/\text{NaCl}$ in, H_2O in, H^+/NH_4^+ out; D: $\text{NaCl}/\text{H}_2\text{O}$ in, HCO_3^- out, K^+/H^+ out (option 2)
- (3) P: HCO_3^- in, NaCl in, H_2O in, H^+/NH_4^+ out; D: H_2O in, HCO_3^- out, NaCl/H^+ out (option 3)
- (4) P: $\text{NaCl}/\text{HCO}_3^-$ in, $\text{H}^+/\text{H}_2\text{O}$ out; D: NaCl in, NH_4^+ out, $\text{K}^+/\text{H}_2\text{O}$ out (option 4)

Q126. Part of bioreactor used as a foam breaker (A=culture port, B=foam breaker outlet, C=impeller area, D=outlet at bottom):

- (1) A
- (2) B
- (3) D
- (4) C

Q127. A: A typical unfertilized angiosperm embryo sac at maturity is 8-nucleate and 7-celled. R: The egg apparatus has 2 polar nuclei.

- (1) A and R true; R is correct explanation
- (2) A and R true; R NOT correct explanation
- (3) A true, R false
- (4) A false, R true

Q128. Specialized membranous structure in prokaryotic cell that helps in cell wall formation, DNA replication and respiration:

- (1) Mesosome
- (2) Chromatophores
- (3) Cristae
- (4) Endoplasmic Reticulum

Q129. Post-transcriptional events in eukaryotes: A. Transport of pre-mRNA to cytoplasm prior to splicing B. Removal of introns and joining of exons C. Methyl group addition at 5' end of hnRNA D. Adenine addition at 3' end E. Base pairing of two complementary RNAs

- (1) A, B, C only
- (2) B, C, D only
- (3) B, C, E only
- (4) C, D, E only

Q130. Pattern of inheritance for polygenic traits:

- (1) Mendelian
- (2) Non-Mendelian
- (3) Autosomal dominant
- (4) X-linked recessive

Q131. Which enzyme contains haem as the prosthetic group?

- (1) RuBisCo
- (2) Carbonic anhydrase
- (3) Succinate dehydrogenase
- (4) Catalase

Q132. Increasing order of body complexity (Whittaker): A. Multicellular heterotrophs (chitin) B. Heterotrophs with tissue/organ/organ-system C. Prokaryotes (polysaccharide + amino acids) D. Eukaryotic autotrophs (tissue/organ) E. Eukaryotes with cellular body organization

- (1) A, C, E, B, D
- (2) C, E, A, D, B
- (3) A, C, E, D, B
- (4) C, E, A, B, D

Q133. Father of Ecology in India:

- (1) S. R. Kashyap
- (2) Ramdeo Misra
- (3) Ram Udar
- (4) Birbal Sahni

Q134. Match: A. Hershey & Chase B. Euchromatin C. Frederick Griffith D. Heterochromatin ↔ I. Streptococcus pneumoniae II. Densely packed, dark-stained III. Loosely packed, light-stained IV. DNA as genetic material confirmation

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-II, C-I, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-II, C-IV, D-I

Q135. Neoplastic cell characteristics: A. Mass of proliferating cells B. Rapid growth C. Invasion and damage to surrounding tissue D. Confined to original location

- (1) A, B only
- (2) A, B, C only
- (3) A, B, D only
- (4) B, C, D only

Q136. Statement I: DNA fragments from gel electrophoresis can be used to make recombinant DNA. Statement II: Smaller fragments observed near anode; larger near wells.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q137. Match: A. Adenosine B. Adenylic acid C. Adenine D. Alanine ↔ I. Nitrogen base II. Nucleotide III. Nucleoside IV. Amino acid

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-III, C-I, D-IV

Q138. Consider: A. Reductive division for human female gametogenesis starts earlier than male. B. Gap between meiotic-I and II is much shorter for males. C. The first polar body is associated with the formation of the primary oocyte. D. LH surge leads to disintegration of the endometrium.

- (1) A and B true
- (2) A and C true
- (3) B and D true
- (4) B and C true

Q139. All living members of class Cyclostomata are:

- (1) Free living
- (2) Endoparasite
- (3) Symbiotic
- (4) Ectoparasite

Q140. A: Primary function of Golgi is to package materials made by ER and deliver them. R: Vesicles from ER fuse with cis face of Golgi; modified and released from trans face.

- (1) Both A and R true; R is correct explanation
- (2) Both A and R true; R is not correct explanation
- (3) A true, R false
- (4) A false, R true

Q141. Match: A. Scutellum B. Non-albuminous seed C. Epiblast D. Perisperm ↔ I. Persistent nucellus II. Cotyledon of monocot seed III. Groundnut IV. Rudimentary cotyledon

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-IV, C-III, D-I

Q142. A: All vertebrates are chordates but all chordates are not vertebrates. R: Members of subphylum Vertebrata possess notochord during embryonic period; replaced by vertebral column in adults.

- (1) Both A and R true; R is correct explanation
- (2) Both A and R true; R is not correct explanation
- (3) A true, R false
- (4) A false, R true

Q143. Which is NOT correct about an antibody?

- (1) Each antibody has 2 light and 2 heavy chains
- (2) Heavy and light chains held by disulfide bonds
- (3) Antigen binding site is at C-terminal
- (4) Constant regions of heavy & light chains are at C-terminus

Q144. Silencing of specific mRNA via RNAi is possible because of:

- (1) Complementary dsRNA
- (2) Inhibitory ssRNA
- (3) Complementary tRNA
- (4) Non-complementary ssRNA

Q145. Genes R and Y assort independently. RRY^Y × rry^y gives F₂ phenotypic ratio:

- (1) 1 : 2 : 1
- (2) 3 : 1
- (3) 9 : 3 : 3 : 1
- (4) 9 : 7

Q146. Histones are enriched with:

- (1) Lysine & Arginine
- (2) Leucine & Lysine
- (3) Phenylalanine & Leucine
- (4) Phenylalanine & Arginine

Q147. First menstruation is called:

- (1) Menopause
- (2) Menarche
- (3) Diapause
- (4) Ovulation

Q148. Match: A. Heart B. Kidney C. Gastro-intestinal tract D. Adrenal Cortex ↔ I. Erythropoietin II. Aldosterone III. ANF IV. Secretin

- (1) A-II, B-I, C-III, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-I, C-IV, D-II

Q149. Protein portion of an enzyme is called:

- (1) Cofactor
- (2) Coenzyme
- (3) Apoenzyme
- (4) Prosthetic group

Q150. Unit of productivity of an ecosystem:

- (1) g m^{-2}
- (2) kCal m^{-2}
- (3) kCal m^{-3}
- (4) $(\text{kCal m}^{-2}) \text{yr}^{-1}$

Q151. Sweet potato and potato represent which evolution?

- (1) Analogy, convergent
- (2) Homology, divergent
- (3) Homology, convergent
- (4) Analogy, divergent

Q152. From the pedigree of an X-linked recessive disease, probability that an F3 child is unaffected and a carrier (carries mutation in one allele):

- (1) $1/4$
- (2) $1/2$
- (3) $1/8$
- (4) Zero

Q153. A: Cells of tapetum possess dense cytoplasm and have more than one nucleus. R: Multiple nuclei in tapetum increase efficiency of nourishing developing microspore mother cells.

- (1) A and R true; R correct explanation
- (2) A and R true; R not correct explanation
- (3) A true, R false
- (4) A false, R true

Q154. Meiotic and mitotic divisions to develop a mature female gametophyte from megaspore mother cell in angiosperm:

- (1) 2 Meiosis and 3 Mitosis
- (2) 1 Meiosis and 2 Mitosis
- (3) 1 Meiosis and 3 Mitosis
- (4) No Meiosis and 2 Mitosis

Q155. Example of a zygomorphic flower:

- (1) Petunia
- (2) Datura
- (3) Pea
- (4) Chilli

Q156. After maturation, lymphocytes migrate to secondary lymphoid organs/tissues like: A. thymus B. bone marrow C. spleen D. lymph nodes E. Peyer's patches

- (1) B, C, D only
- (2) A, B, C only
- (3) E, A, B only
- (4) C, D, E only

Q157. Statement I: Fig fruit is non-vegetarian as it has enclosed fig wasps. Statement II: Fig wasp and fig tree show mutualism.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q158. Main function of spindle fibers during mitosis:

- (1) Separate chromosomes
- (2) Synthesize new DNA
- (3) Repair damaged DNA
- (4) Regulate cell growth

Q159. Characteristic feature of gymnosperms:

- (1) Seeds enclosed in fruits
- (2) Seeds are naked
- (3) Seeds are absent
- (4) Have flowers for reproduction

Q160. Functions of adrenal medullary hormones: A. Pupillary constriction B. Hyperglycemic C. Piloerection D. Increases strength of heart contraction

- (1) C and D only
- (2) B, C and D only
- (3) A, C and D only
- (4) D only

Q161. Why is insulin not given orally to diabetics?

- (1) Strong immune response
- (2) Digested in GI tract
- (3) Structural variation
- (4) Bioavailability increases

Q162. Match: A. Pteridophyte B. Bryophyte C. Angiosperm D. Gymnosperm ↔ I. Salvia II. Ginkgo III. Polytrichum IV. Salvinia

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

Q163. Who proposed that the genetic code for amino acids is made of three nucleotides?

- (1) George Gamow
- (2) Francis Crick
- (3) Jacques Monod
- (4) Franklin Stahl

Q164. Match: A. The Evil Quartet B. Ex situ conservation C. Lantana camara D. Dodo ↔ I. Cryopreservation II. Alien species invasion III. Causes of biodiversity loss IV. Extinction

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-IV, D-I

Q165. Hormone released from the pituitary but synthesized in the hypothalamus:

- (1) LH
- (2) ADH
- (3) FSH
- (4) ACTH

Q166. Role of water vascular system in Echinoderms: A. Respiration & Locomotion B. Excretion & Locomotion C. Capture & transport of food D. Digestion & Respiration E. Digestion & Excretion

- (1) A and B only
- (2) A and C only
- (3) B and C only
- (4) B, D and E only

Q167. Type of immunity present at birth and non-specific:

- (1) Acquired
- (2) Innate
- (3) Cell-mediated
- (4) Humoral

Q168. In bryophytes, gemmae help in:

- (1) Sexual reproduction
- (2) Asexual reproduction
- (3) Nutrient absorption
- (4) Gaseous exchange

Q169. In frog, the renal portal system links:

- (1) Liver and intestine
- (2) Liver and kidney
- (3) Kidney and intestine
- (4) Kidney and lower part of body

Q170. Statement I: In ecosystems, energy flow from producers to consumers is unidirectional. Statement II: Ecosystems are exempted from the 2nd law of thermodynamics.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q171. Which is true about RuBisCO?

- (1) Active only in dark
- (2) Higher affinity for O_2 than CO_2
- (3) Involved in photolysis of water
- (4) Catalyzes carboxylation of RuBP

Q172. Which enzyme(s) is/are NOT essential for gene cloning? A. Restriction enzymes B. DNA ligase C. DNA mutase D. DNA recombinase E. DNA polymerase

- (1) C and D only
- (2) A and B only
- (3) D and E only
- (4) B and C only

Q173. True statements about plant growth: A. Parthenocarp can be induced by auxins. B. PGRs can promote and inhibit growth. C. Dedifferentiation is prerequisite for redifferentiation. D. ABA is a growth promoter. E. Apical dominance promotes lateral bud growth.

- (1) A, B, C only
- (2) A, C, E only
- (3) A, D, E only
- (4) B, D, E only

Q174. Factor important for termination of transcription:

- (1) α (alpha)
- (2) σ (sigma)
- (3) ρ (rho)
- (4) γ (gamma)

Q175. Frogs respire in water by skin and buccal cavity, and on land by skin, buccal cavity and lungs.

- (1) True for water, false for land
- (2) True for both
- (3) False for water, true for land
- (4) False for both

Q176. Twins (one boy, one girl) born to a family. Which must be true?

- (1) Monozygotic twins
- (2) Fraternal twins
- (3) Conceived through IVF
- (4) 75% identical genetic content

Q177. Which microbes are NOT involved in preparation of household products? A. *Aspergillus niger* B. *Lactobacillus* C. *Trichoderma polysporum* D. *Saccharomyces cerevisiae* E. *Propionibacterium sharmanii*

- (1) A and B only
- (2) A and C only
- (3) C and D only
- (4) C and E only

Q178. Match: A. Progesterone B. Relaxin C. Melanocyte stimulating hormone D. Catecholamines $\leftrightarrow$ I. Pars intermedia II. Ovary III. Adrenal Medulla IV. Corpus luteum

- (1) A-IV, B-II, C-I, D-III
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

Q179. Blue/white selectable markers: Statement I: Blue colonies have DNA insert and are recombinant. Statement II: Colonies without blue colour have insert and are recombinant.

- (1) Both I and II correct
- (2) Both I and II incorrect
- (3) I correct, II incorrect
- (4) I incorrect, II correct

Q180. Verhulst-Pearl Logistic Growth equation:

- (1) $dN/dt = r((K - N)/K)$
- (2) $dN/dt = rN((K - N)/K)$
- (3) $dN/dt = rN((N - K)/N)$
- (4) $dN/dt = N((r - K)/K)$

Official Answer Key — NEET (UG) 2025

All answers are referenced to **Test Booklet Code 45** as released by the conducting authority.

Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans
1	(2)	46	(4)	91	(2)	136	(1)
2	(2)	47	(2)	92	(2)	137	(3)
3	(2)	48	(2)	93	(1)	138	(1)
4	(4)	49	(1)	94	(4)	139	(4)
5	(2)	50	(1)	95	(1)	140	(1)
6	(2)	51	(3)	96	(1)	141	(1)
7	(4)	52	(3)	97	(1)	142	(1)
8	(4)	53	(2)	98	(1)	143	(3)
9	(1)	54	(4)	99	(3)	144	(1)
10	(1)	55	(1)	100	(3)	145	(3)
11	(4)	56	(4)	101	(4)	146	(1)
12	(1)	57	(2)	102	(2)	147	(2)
13	(3)	58	(4)	103	(3)	148	(4)
14	(3)	59	(4)	104	(4)	149	(3)
15	(2)	60	(4)	105	(4)	150	(4)
16	(2)	61	(3)	106	(3)	151	(1)
17	(4)	62	(2)	107	(4)	152	(1)
18	(2)	63	(1)	108	(4)	153	(3)
19	(3)	64	(1)	109	(2)	154	(3)
20	(1)	65	(1)	110	(1)	155	(3)
21	(4)	66	(2)	111	(1)	156	(4)
22	(2)	67	(3)	112	(1)	157	(2)
23	(2)	68	(2)	113	(2)	158	(1)
24	(3)	69	(2)	114	(4)	159	(2)
25	(3)	70	(3)	115	(3)	160	(2)
26	(3)	71	(3)	116	(1)	161	(2)
27	(1)	72	(1)	117	(3)	162	(2)
28	(4)	73	(3)	118	(1)	163	(1)
29	(2)	74	(1)	119	(3)	164	(2)
30	(3)	75	(4)	120	(4)	165	(2)
31	(1)	76	(1)	121	(4)	166	(2)
32	(2)	77	(1)	122	(2)	167	(2)

33	(3)	78	(2)	123	(4)	168	(2)
34	(2)	79	(3)	124	(4)	169	(4)
35	(2)	80	(3)	125	(2)	170	(3)
36	(2)	81	(2)	126	(4)	171	(4)
37	(3)	82	(2)	127	(3)	172	(1)
38	(1)	83	(2)	128	(1)	173	(1)
39	(3)	84	(1)	129	(2)	174	(3)
40	(2)	85	(1)	130	(2)	175	(3)
41	(2)	86	(3)	131	(4)	176	(2)
42	(3)	87	(1)	132	(2)	177	(2)
43	(3)	88	(4)	133	(2)	178	(1)
44	(4)	89	(1)	134	(3)	179	(4)
45	(2)	90	(4)	135	(2)	180	(2)

Marking Scheme: Correct: +4 | Incorrect: -1 | Unattempted: 0

Note: For Q63, both options (1) and (2) were considered correct as per the official key.